ALICE L WALTON SCHOOL OF MEDICINE

Principles of Antibacterial Drugs

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Adapted from: Antibacterial Drugs: Foundations and Frameworks

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Learning Objectives (2) Versions

After completing this brick, you will be able to:

1. Explain the relevance of Minimum Inhibitory Concentration (MIC) and Minimum Bactericidal Concentration (MBC) in antibiotic therapy.
2. Use susceptibility testing, antibiograms, and local resistance patterns to select appropriate therapeutic agents.

CASE CONNECTION

MG is a 55-year-old female admitted to the hospital for a severe infection two days ago. She has a history of immunosuppression due to chronic medication therapy for her rheumatoid arthritis. Initially, her condition improved, however, it does not seem that the infection is resolving completely like it should.

Empiric antibiotics were started when she arrived. Today you get the culture and sensitivity report from the microbiology lab. It shows that the antibiotic that was started initially has '*intermediate*' coverage against the bacteria in her blood.

In the context of MIC, what does '*intermediate*' mean? Knowing that MG is immunosuppressed, would it change the type of antibiotic you would use? Consider your answers to these questions as you work through this brick and we'll revisit MG at the end.

GO TO CONCLUSION ↓

As you will see throughout this course there are many different antibacterial agents. It can be difficult to know which one to choose, and as bacteria grow and evolve and in different regions of the country and world bacterial resistance patterns can alter the antibiotic therapy you would choose. In this brick we will focus on some guiding principles like minimum inhibitory concentration (MIC), minimum bactericidal concentration (MBC), susceptibility, and antibiograms. As we discuss these, and as you learn about some of the antibacterial medications in this course, it will help inform your decisions on which therapy will work best to treat bacterial infections.

What are MIC and MBC? (LO#1)

In clinical and hospital practice, a measurement called the minimum inhibitory concentration (MIC) determines antibiotic susceptibility. MIC, as the name suggests, is the minimal antibiotic concentration required to inhibit the growth of the bacterium in question. This is different from the minimum bactericidal concentration (MBC), which is the minimum concentration of antibiotic that would result in bacterial cell death (defined as inability to re-culture the bacteria). MBC is rarely used in practice, but it is used to determine if an antibacterial is more bactericidal than bacteriostatic.

CLINICAL CORRELATION

MICs can often be easily calculated through assays that are commercially available, many of which can return results in about 24 hours. MBC, however, is much more difficult to obtain and has little clinical value given the efficacy of bacteriostatic action. So it is rare for MBC data to be acquired for patients outside of research purposes.

So how is the MIC determined? The classic method, shown in figure 1, was done in test tubes, now it is done on microdilution plates, but the concept is the same. The microbiology laboratory determines the MIC by mixing standard concentration of the organism (from a patient sample, or "culture") with increasing concentrations of the antibiotic. Note the increasing concentrations in µg/mL below the test tubes in the figure. The culture/ antibiotic mixtures are incubated for about a [illegible]ned (with the naked eye and computer aided) for signs of cloudiness, this indicates growth of the bacterial organism. The mixture with the lowest concentration of antibiotic where there is no visible growth is deemed the MIC. For each organism-antibiotic pair there is a particular cutoff MIC that is considered '*susceptible*'. This particular MIC is called the breakpoint, it is closely monitored and published by a national organization. In clinical practice, the microbiology laboratory will provide reports that help guide antibiotic therapy. We will take a closer look at culture and sensitivity reports in the next section.

How is the MBC determined? In the same type of dilution testing as shown in figure 1 for MIC the MBC can be determined. The concentration of antibiotic that corresponds to a 99.9% reduction in the original bacterial colony count is considered the MBC.

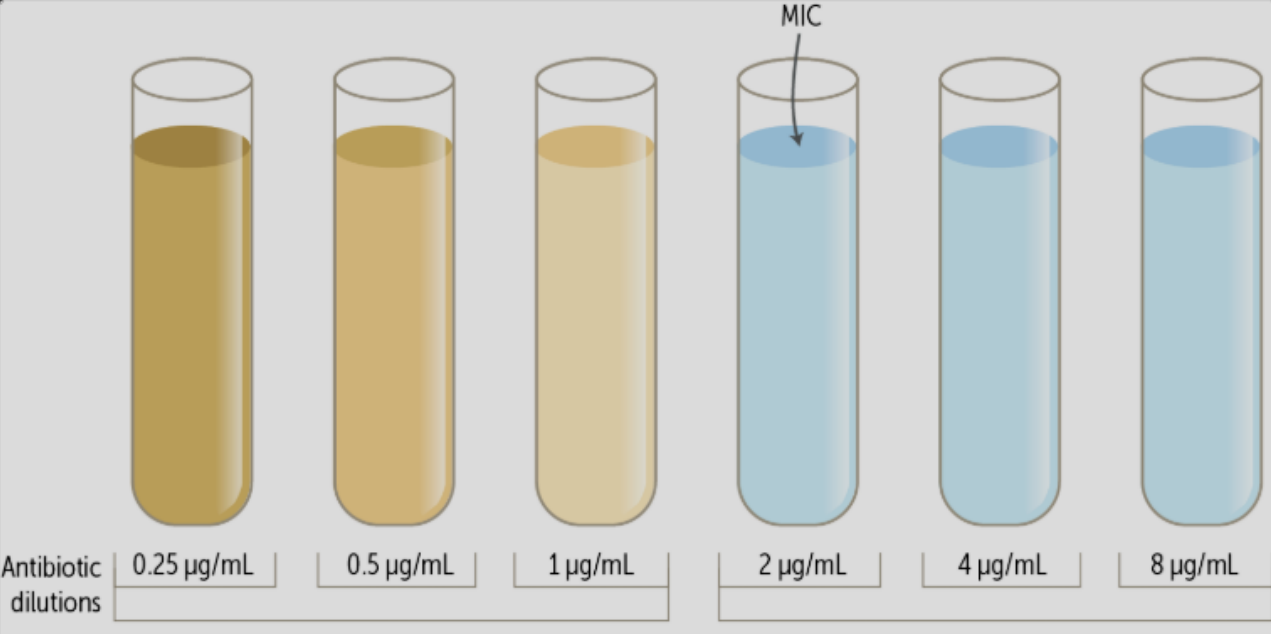

MIC
Antibiotic dilutions
0.25 µg/mL
0.5 µg/mL
1 µg/mL
2 µg/mL
4 µg/mL
8 µg/mL
Turbidity = bacterial growth
No turbidity = inhibition of bacterial growth

Figure 1

Note, ***just because an antibiotic has the lowest MIC for a pathogen does not mean it is the best choice*** - different antibiotics achieve different concentrations in the body in different places (e.g. CNS vs bladder). So, antibiotics MICs for a single organism generally should not be compared across different different drugs in selecting therapy.

So, what does this all mean? Well, bacteriostatic antibacterials are agents that halt the growth of bacteria. When bacterial growth and replication are halted, the immunocompetent host is able to more easily clear the organism—making these agents as effective as bactericidal agents. As the MIC approaches the MBC, the agent is considered more bactericidal. Bactericidal antibacterials are those that actually kill bacteria as their primary action. Whether a drug is considered bacteriostatic or bactericidal, depends on a comparison of the MIC to MBC as shown in Table 1. Table 2 shows a list of common antibiotics (which will be covered throughout the course) and whether they are bacteriostatic or -cidal.

Table 1

Bactericidal	MBC/MIC ≤ 4
Bacteriostatic	MBC/MIC > 4

For most infections, outcomes using appropriate bacteriostatic versus bactericidal drugs are similar; however, for certain infections bactericidal drugs are preferred (e.g., endocarditis, meningitis). In some cases, the immune system may not be as [illegible] because of the anatomic location, or the immunosuppression of the patient.

T > MIC

Besides differing in whether they kill microorganisms or merely inhibit their growth, antibiotics also differ in how they manifest their effects over time. Studies have shown that for certain antibiotics, activity against microorganisms correlates with the *duration of time that the concentration of drug remains above the MIC*. This is referred to as time-dependent activity, or antibiotics that are time dependent. Practically, antibiotics in this category will require more frequent dosing (e.g., three or four times daily).

Cmax / MIC

For some antibiotics the ratio of the *peak concentration of the drug over the MIC* determines the antibiotic efficacy. These antibiotics are referred to as *concentration-dependent* antibiotics. Clinically, these antibiotics are often only dosed once daily and rely on larger doses to maintain their concentration over the MIC.

AUC / MIC

Lastly, for some antibiotics the best predictor of activity is the ratio of the area under the concentration-time curve (AUC) to the MIC. All of these principles are demonstrated in Figure 2 below.

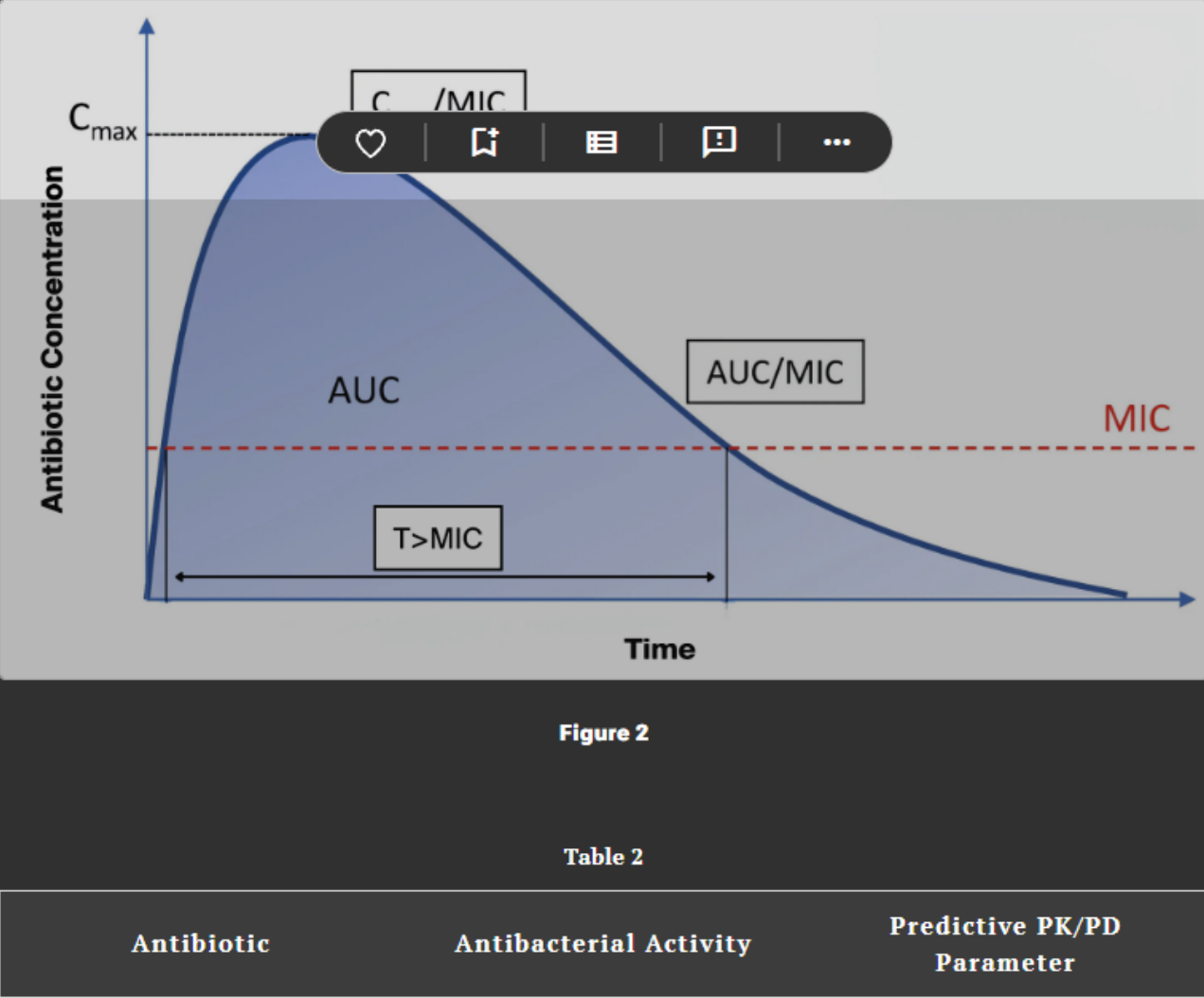


Figure 2

Table 2

Antibiotic	Antibacterial Activity	Predictive PK/PD Parameter
Penicillins Cephalosporins Carbapenems Monobactams	Bactericidal	Time > MIC
Vancomycin	Bactericidal (slow)	AUC / MIC
Fluoroquinolones Aminoglycosides Metronidazole Daptomycin	Bactericidal	Cmax / MIC
Macrolides Tetracyclines Linezolid	Bacteriostatic	AUC / MIC

What is time-dependent activity in antibiotics?

How Do We Treat Bacterial Infections While Waiting for Test Results? (LO#2)

As mentioned in the section above, MIC, or sensitivity testing, takes about 24 hours, or longer. While we wait for cultures to identify the cause of an infection, we may not have the time to wait on starting treatment. After all, in some cases, minutes really do matter (e.g., in acute bacterial meningitis), and some cultures can take days to return with species identification and/ or susceptibility results. In these cases, ***empiric therapy*** ("best guess") is started to cover the likely (and most dangerous) culprits. Why? We can always de-escalate treatments, but if we do not properly cover patients, not only do we not adequately treat the infection, but we are also exposing them to an antibacterial that may have its own side effects without actually treating the infection. Making the [illegible]n be difficult. It requires an understanding of the *patient's disease process and potential sources of infection*, considering the *most common pathogens* (i.e., normal flora) based on the source of infection, and *local resistance rates*.

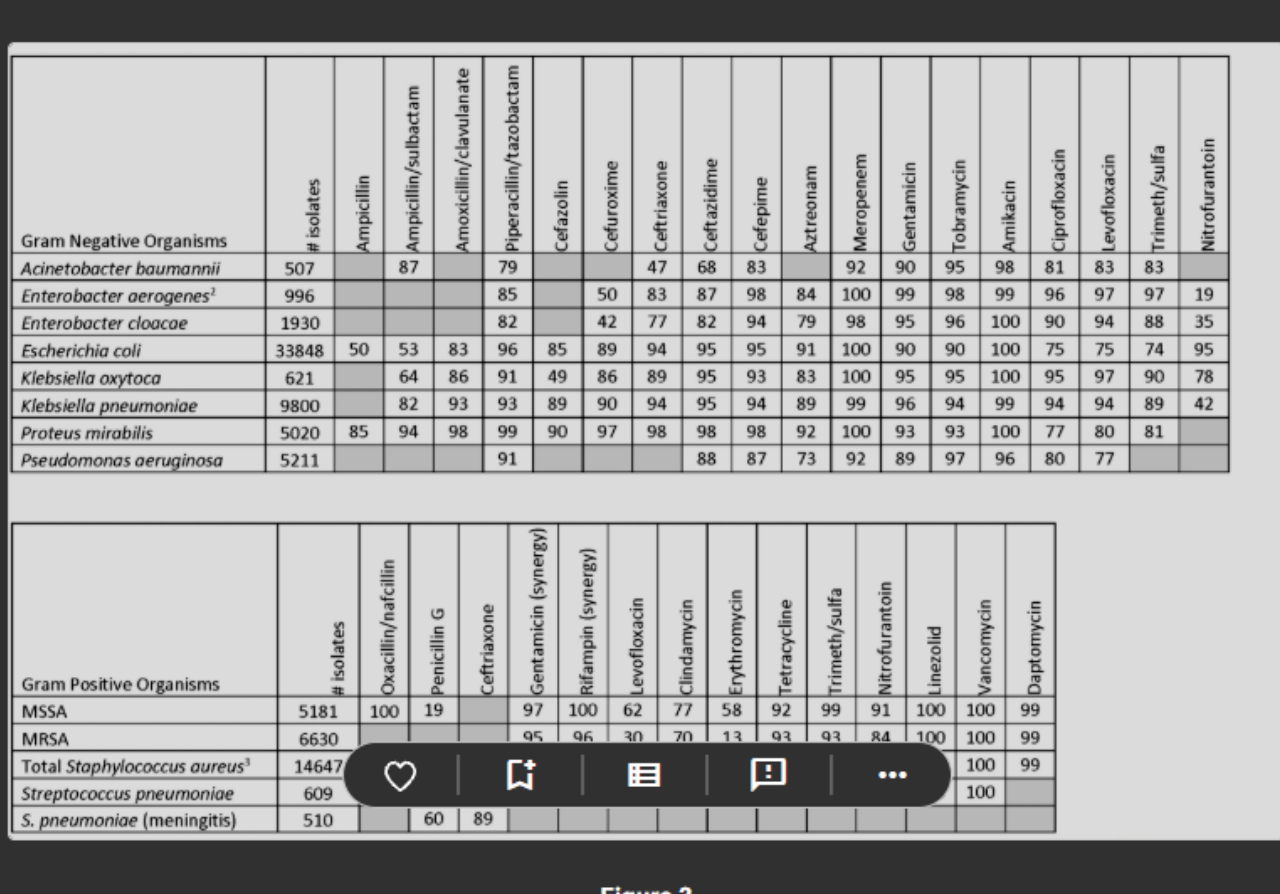
How do you choose empiric therapy?

To adequately consider empiric treatments, ***local surveillance*** (antibiotic susceptibility monitoring) data and knowledge of disease epidemiology and occurrence of common pathogen(s) are crucial. Different regions have variations in bacterial resistance rates and infection rates. All culture and sensitivity reports (described below) are recorded and reported. Then, an ***antibiogram*** can be created. An antibiogram can be specific to a healthcare facility, city, county, region, state, country, and globally. Various organizations, generally the Centers for Disease Control and prevention (CDC), and local health departments maintain antibiograms. *Antibiograms are a crucial tool in choosing empiric antibiotic therapy* as variation in resistance can vary widely between areas.

What is an antibiogram?

An antibiogram is a profile of how susceptible a specific microorganism is to a range of antimicrobial drugs. An example is shown below in Figure 3. It's usually specific to a healthcare facility and is based on aggregate data from a hospital or healthcare system. Antibiograms include:

- Percentages of organisms tested that are susceptible to a particular antimicrobial drug.
- Data from a defined time frame (usually one year or six months).
- Information about which antimicrobials are more or less effective for different organisms.



Gram Negative Organisms	# isolates	Ampicillin	Ampicillin/sulbactam	Amoxicillin/clavulanate	Piperacillin/tazobactam	Cefazolin	Cefuroxime	Ceftriaxone	Ceftazidime	Cefepime	Aztreonam	Meropenem	Gentamicin	Tobramycin	Amikacin	Ciprofloxacin	Levofloxacin	Trimeth/sulfa	Nitrofurantoin
Acinetobacter baumannii	507		87		79			47	68	83		92	90	95	98	81	83	83	
Enterobacter aerogenes[2]	996				85		50	83	87	98	84	100	99	98	99	96	97	97	19
Enterobacter cloacae	1930				82		42	77	82	94	79	98	95	96	100	90	94	88	35
Escherichia coli	33848	50	53	83	96	85	89	94	95	95	91	100	90	90	100	75	75	74	95
Klebsiella oxytoca	621		64	86	91	49	86	89	95	93	83	100	95	95	100	95	97	90	78
Klebsiella pneumoniae	9800		82	93	93	89	90	94	95	94	89	99	96	94	99	94	94	89	42
Proteus mirabilis	5020	85	94	98	99	90	97	98	98	98	92	100	93	93	100	77	80	81	
Pseudomonas aeruginosa	5211				91				88	87	73	92	89	97	96	80	77		

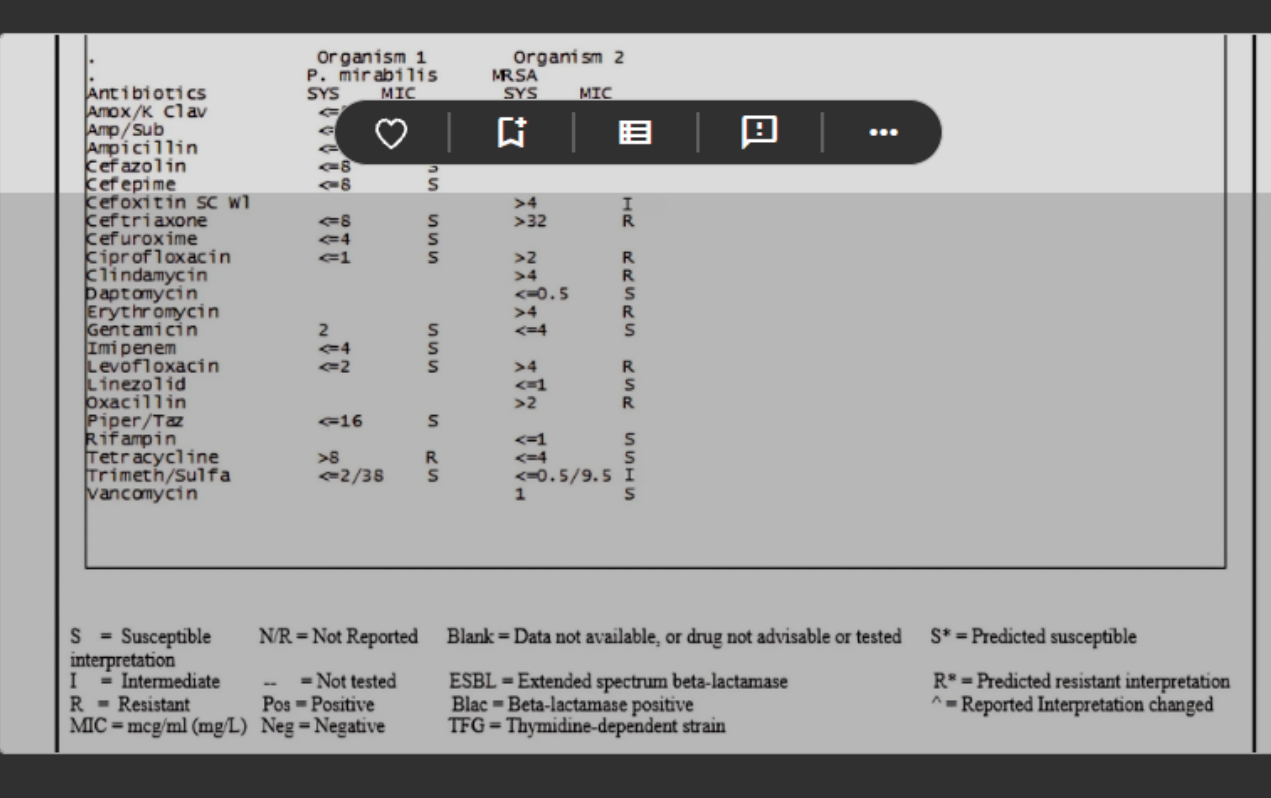
Gram Positive Organisms	# isolates	Oxacillin/nafcillin	Penicillin G	Ceftriaxone	Gentamicin (synergy)	Rifampin (synergy)	Levofloxacin	Clindamycin	Erythromycin	Tetracycline	Trimeth/sulfa	Nitrofurantoin	Linezolid	Vancomycin	Daptomycin
MSSA	5181	100	19		97	100	62	77	58	92	99	91	100	100	99
MRSA	6630				95	96	30	70	13	93	93	84	100	100	99
Total *Staphylococcus aureus*[3]	14647	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	100	99
Streptococcus pneumoniae	609	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	100	
S. pneumoniae (meningitis)	510		60	89											

Figure 3

CLINICAL CORRELATION

Many hospitals and health departments keep track of the development of resistant strains given their massive public health impact. The development of new multidrug-resistant organisms, for example, is often a nationally or even globally newsworthy event, such as the confirmation of highly resistant *Neisseria gonorrhoeae* strains.

As we talked about in the section above, the microbiology lab completes MIC testing based on a patient sample. Then they provide a ***culture and sensitivity*** report that can be used clinically. An example of a culture and sensitivity report is shown in Figure 4. This report has the organism identified with a list of antibiotics and whether organism is ***susceptible (S)***, ***intermediate (I)***, or ***resistant (R)***. The culture and sensitivity report provides you, the clinician, with patient specific information on the bacteria they are infected with and which antibiotic(s) will treat the infection, based on their MIC testing. Then, using your clinical judgement, you determine the best antibiotic choice for the patient. Compared to the best guess, empiric therapy, you will determine ***definitive therapy***.



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                 Organism 1        Organism 2
                 P. mirabilis      MRSA
Antibiotics      SYS    MIC        SYS    MIC
Amox/K Clav      <=
Amp/Sub          <
Ampicillin       <
Cefazolin        <=8    S
Cefepime         <=8    S
Cefoxitin SC W1                    >4     I
Ceftriaxone      <=8    S          >32    R
Cefuroxime       <=4    S
Ciprofloxacin    <=1    S          >2     R
Clindamycin                        >4     R
Daptomycin                         <=0.5  S
Erythromycin                       >4     R
Gentamicin       2      S          <=4    S
Imipenem         <=4    S
Levofloxacin     <=2    S          >4     R
Linezolid                          <=1    S
Oxacillin                          >2     R
Piper/Taz        <=16   S
Rifampin                           <=1    S
Tetracycline     >8     R          <=4    S
Trimeth/Sulfa    <=2/38 S          <=0.5/9.5 I
Vancomycin                         1      S
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S = Susceptible interpretation
I = Intermediate
R = Resistant
MIC = mcg/ml (mg/L)

N/R = Not Reported
-- = Not tested
Pos = Positive
Neg = Negative

Blank = Data not available, or drug not advisable or tested
ESBL = Extended spectrum beta-lactamase
Blac = Beta-lactamase positive
TFG = Thymidine-dependent strain

S* = Predicted susceptible
R* = Predicted resistant interpretation
^ = Reported Interpretation changed

Figure 4

Putting all of these principles from this brick together, let's think about a possible scenario.

Imagine a patient coming to the clinic with a thigh abscess. Although this would probably be a simple infection in some patients, this patient is a known intravenous drug user who shares needles and has had multiple emergency department visits in the past month. This significantly raises our concern for a resistant organism, mostly MRSA, and the local *antibiogram* shows resistance rates to specific antibiotics. Both of these factors would play a role in determining your chosen *empiric therapy*. As you further evaluate the patient you are able to get a sample of the wound and you send it to the microbiology lab for *culture and sensitivity*. You send the patient home with a course of antibiotics. In the meantime, in the lab they identify the organism and perform *MIC testing* to determine susceptibility. A *culture and sensitivity report* is prepared by the lab a[illegible]sistant organism, rather a common gram-positive pathogen that is *susceptible* to multiple antibiotics. You are able to call the patient and de-escalate their *definitive therapy* to an oral *time-dependent bacteriostatic* agent.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to MG, she has a bacterial infection that is not completely resolving. This could be for a couple of reasons. First, since she is immunosuppressed the type of antibiotic might play a role, meaning bacteriostatic vs -cidal. Often, for these patients bactericidal agents are preferred because the immune system cannot take care of the pathogen once the bacteriostatic antibiotic has halted growth. Additionally, based on the culture and sensitivity report the antibiotic has *intermediate* coverage against the bacteria. This means that choosing another antibiotic, to which the bacteria is *susceptible*, would be out best course of action.

Summary

- Minimum inhibitory concentration (MIC) is the lowest concentration of an antibiotic that inhibits bacterial growth, while minimum bactericidal concentration (MBC) is the lowest concentration that kills bacteria.
- MIC is commonly used in clinical practice to determine antibiotic susceptibility.
- Bacteriostatic antibiotics inhibit bacterial growth, allowing the immune system to clear the infection, while bactericidal antibiotics kill bacteria directly.
- Antibiotics can b[illegible]endent, or AUC/MIC-dependent, influencing dosing frequency and effectiveness.
- Empiric therapy is initiated before culture results are available, using local antibiograms and knowledge of common pathogens to guide antibiotic choice.
- Culture and sensitivity reports provide specific information on bacterial susceptibility, allowing clinicians to adjust from empiric to definitive therapy.

Review Questions

1. What determines if an antibiotic is bacteriostatic or bactericidal?

A. ● MIC to MBC comparison
B. ○ MIC
C. ○ MBC
D. ○ Antibiotic concentration in the CNS
E. ○ Pathogen resistance

Hide Explanation

The correct answer is MIC to MBC comparison (A). This comparison determines whether an antibiotic halts bacterial growth (bacteriostatic) or kills bacteria (bactericidal). The lowest MIC (B) alone does not determine this, as different antibiotics achieve di[illegible]ne (C) is not sufficient without comparing it to MIC. Antibiotic concentration in the CNS (D) is relevant for treatment efficacy but not for determining bacteriostatic vs. bactericidal. Pathogen resistance (E) affects treatment choice but not the classification of the antibiotic.

2. What does an antibiogram show?

A. ● Susceptibility of microorganisms
B. ○ Number of infections
C. ○ Types of microorganisms
D. ○ Hospital infection rates
E. ○ Patient treatment outcomes

Hide Explanation

The correct answer is susceptibility of microorganisms (A). An antibiogram provides data on how susceptible different microorganisms are to various antimicrobial drugs. It does not show the number of infections (B), types of microorganisms (C), hospital infection rates (D), or patient treatment outcomes (E).

How would you rate this Brick?

Hide Explanation

The correct answer is susceptibility of microorganisms (A). An antibiogram provides data on how susceptible different microorganisms are to various antimicrobial drugs. It does not show the number of infections (B), types of microorganisms (C), hospital infection rates (D), or patient treatment outcomes (E).

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the relevance of Minimum Inhibitory Concentration (MIC) and Minimum Bactericidal Concentration (MBC) in antibiotic therapy.~~

02 ☑ ~~Use susceptibility testing, antibiograms, and local resistance patterns to select appropriate therapeutic agents.~~

Objective Confidence: 2/2